

```
plotted = Button(w, text = 'PLOT THE FIELD LINES', command
= Plotter).pack(pady = 10)
```

How to use the code

Make sure that the Tkinter module is available in the IDE being used. If it is not, you can download it, based on your operating system. Once the GUI opens on running the code, the details are to be entered independently—like the position of the charge and the magnitude for each charge; and the *add charge details* button should be pressed. This should be done for all the charges you want, one by one, and after successively adding charges when the *plot field lines* button is pressed, the field lines are plotted for the set of charges.

For example, for a system of four charges, the steps involved in obtaining the field lines can be seen from Figures 2 to 5. In Figure 2, the first charge has (-2, 0, -2) as its x-coordinate, y-coordinate and charge magnitude. In Figure 3, the second charge has (2, 0, 2) as its x-coordinate, y-coordinate and charge magnitude. The third and fourth charges in Figures 4 and 5 have

(0, 2, -1) and (0, -2, 1), respectively, as their x-coordinate, y-coordinate and charge magnitude.

The plot of field lines obtained for this system of charges can be seen in Figure 6. For any system of static charges, field lines can be plotted in this manner.

The code shown isn't perfect and can always be made more efficient based on one's needs. **END** 

References

- [1] Python documentation for all the modules used from www.python.org
- [2] The syntax for defining the class 'charge' was from <https://scipython.com/blog/visualizing-a-vector-field-with-matplotlib/>
- [3] Tkinter documentation from www.python.org

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suggestion or a complaint?**

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